SIH26054 — Complete Project Brief

AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs

1. Problem Statement

MALE UAVs depend on piston engines for long-duration missions, where engine reliability is critical. Conventional maintenance may rely on fixed schedules and post-flight inspection, making it difficult to identify developing faults early. The proposed system creates a digital twin of the UAV piston engine that continuously represents its operating condition and uses engineering models and AI/ML to detect abnormal behaviour, predict faults, track degradation and estimate Remaining Useful Life (RUL).

2. Proposed Solution

Engine / Simulated Engine → Sensor Data → Digital Twin → AI/ML Analysis → Operator Dashboard

The prototype can use physics/engineering-based simulation and synthetic or public data instead of a physical UAV engine. It monitors parameters such as RPM, temperature, pressure, vibration, fuel flow and engine load.

3. End-to-End Workflow

Stage	What happens
1. Engine configuration	Define engine characteristics, operating limits, mission duration and operating conditions.
2. Data acquisition	Receive simulated, public or future real-time sensor data: RPM, temperature, pressure, vibration, fuel flow and load.
3. Data processing	Validate readings, filter noise, handle missing values, normalize data and create time-series inputs.
4. Digital twin	Update the virtual engine state and compare expected engine behaviour with observed behaviour.
5. Health monitoring	Calculate an engine-health indicator from current parameters and degradation trends.
6. Anomaly detection	Identify behaviour that deviates from expected operating patterns.
7. Fault prediction	Classify likely faults from abnormal parameter combinations and trends.
8. Degradation tracking	Track how engine condition changes over operating time.
9. RUL estimation	Estimate remaining useful operating life from degradation history.
10. Mission simulation	Simulate takeoff, climb, cruise, high-load operation, descent and landing.
11. Mission replay	Replay parameter and health trends to show when anomalies and degradation developed.
12. Dashboard	Display live parameters, health, alerts, predicted faults, RUL and historical mission information.

4. Example Fault Workflow

Healthy: normal parameters → high health → no alert.
Gradual degradation: vibration increases + temperature changes + efficiency decreases → anomaly detected → health falls.
Serious fault: multiple abnormal parameters → likely fault identified → severity warning → RUL updated.

5. Digital Twin

The digital twin is the virtual representation of the engine. It combines current sensor values with an engineering model to estimate engine state. The key concept is comparing expected behaviour with actual behaviour rather than merely displaying raw sensor numbers.

6. AI/ML Layer

Function	Possible approach
Anomaly detection	Isolation Forest, Autoencoder or another suitable anomaly detector
Fault classification	Random Forest, XGBoost or another supervised classifier
Health estimation	Regression or a data-driven health model
Degradation tracking	Time-series analysis/model
RUL prediction	Regression or a suitable time-series model such as LSTM/GRU if justified

7. Operator Dashboard

The dashboard should present engine health, live sensor values, anomaly/fault alerts, predicted RUL, parameter graphs, historical data and mission replay. A representative display might show Engine Health 78%, RUL 137 hours, RPM 4,320, temperature

218°C, oil pressure 4.1 bar, vibration 0.42 g, and a possible fault with model confidence.

8. Mission Simulation

Takeoff → **Climb** → **Cruise** → **High-load operation** → **Descent** → **Landing.** Each phase changes engine operating conditions. Fault injection demonstrates how the digital twin and ML system respond.

9. Mission Replay & Reporting

After a mission, replay RPM, temperature, vibration, health and other parameters over time, identify when degradation began, see when the system raised an alert, and generate a maintenance-oriented report.

10. Team Roles

Role	Primary responsibility
Engine / Domain Research	Engine physics, parameters, common faults, research papers and validation ranges.
Digital Twin / Simulation	Physics/engineering model, engine state, operating conditions and fault simulation.
ML / Data Science	Anomaly detection, fault classification, degradation modelling and RUL estimation.
Backend / Data Pipeline	APIs, database, data ingestion, model integration and system communication.
Frontend / Dashboard	Real-time UI, charts, engine visualisation, alerts, mission replay and reports.
Integration / Testing	System integration, test scenarios, validation, edge cases, documentation and demo preparation.

11. Development Plan

Phase 1 — Research: piston-engine operation, parameters, faults, digital twins, predictive maintenance and RUL.
Phase 2 — Simulation: representative engine model and realistic time-series data.
Phase 3 — Data Pipeline: store, validate and preprocess sensor data.
Phase 4 — Digital Twin: virtual engine state connected to incoming data.
Phase 5 — ML: anomaly detection, then fault classification, degradation and RUL.
Phase 6 — Backend: connect simulation, database and ML through APIs.
Phase 7 — Dashboard: operator interface and mission replay.
Phase 8 — Integration & Testing: healthy operation, degradation, faults, noise, missing data and abnormal events.

12. Recommended MVP Scope

For a first-year team: **one representative piston engine + realistic simulated sensor data + digital twin + anomaly detection + a small set of fault classes + basic degradation/RUL estimation + polished dashboard.**

Do not initially attempt a perfect CFD simulation, real UAV-engine integration, every possible fault, or a highly complex deep-learning system. Add those only after the core system works reliably.

13. Final Objective

The final prototype should demonstrate a virtual UAV engine that monitors itself, identifies abnormal behaviour, predicts developing faults, estimates remaining useful life and communicates actionable information to an operator in real time.